

Customer Churn Analysis Report

1. Project Title

Customer Churn Analysis Using Exploratory Data Analysis (EDA)

2. Objective

The objective of this project is to analyze customer churn data and identify the factors that influence customer attrition. Exploratory Data Analysis (EDA) was performed to understand customer behavior, detect patterns, identify outliers, and discover relationships between customer characteristics and churn status.

3. Dataset Information

Dataset Name

Customer Churn Dataset

Dataset Source

Kaggle / Customer Churn Dataset

Dataset Description

The dataset contains customer-level information such as demographic details, account information, service usage, and churn status. Each row represents an individual customer.

Dataset Summary

Attribute	Value
Number of Rows	After Cleaning: _____
Number of Columns	_____
Target Variable	Churn
Numerical Features	Tenure, Monthly Charges, Total Charges, etc.
Categorical Features	Gender, Contract Type, Payment Method, etc.

4. Data Cleaning Process

The following data cleaning techniques were performed:

Missing Values

- Checked all columns for missing values.
- Numerical columns were filled using the median value.

- Categorical columns were filled using the mode value.

Duplicate Records

- Duplicate rows were identified and removed.
- This ensured that each customer record appeared only once.

Data Type Validation

- Data types of all columns were reviewed.
- Columns were converted to appropriate formats where required.

Removal of Unnecessary Columns

- Customer ID and other identifier columns were removed because they do not contribute to churn prediction.

Inconsistent Values

- Extra spaces and formatting issues in categorical columns were cleaned.

Outlier Detection

- Outliers were identified using the Interquartile Range (IQR) method.
- Boxplots were used to visually inspect extreme values.

5. Exploratory Data Analysis (EDA)

5.1 Univariate Analysis

Univariate analysis was performed to study individual variables.

Numerical Features

- Histograms were used to visualize distributions.
- Mean, median, minimum, maximum, and standard deviation were calculated.

Categorical Features

- Frequency counts were generated.
- Countplots were used to understand category distributions.

5.2 Bivariate Analysis

Bivariate analysis examined the relationship between customer churn and other variables.

Numerical Variables vs Churn

- Boxplots were used to compare distributions across churn groups.

- Differences in tenure, charges, and usage patterns were observed.

Categorical Variables vs Churn

- Countplots with churn categories were created.
- Contract type, payment method, and customer demographics showed varying churn rates.

5.3 Correlation Analysis

- A correlation heatmap was generated for numerical variables.
- Strong positive and negative relationships among features were identified.
- Highly correlated features were noted for future modeling purposes.

5.4 Outlier Analysis

- Boxplots highlighted unusual observations.
- Most numerical features contained some outliers.
- Outliers were retained because they may represent genuine customer behavior.

6. Visualizations Generated

The following visualizations were created:

1. Customer Churn Count Plot
2. Customer Churn Percentage Pie Chart
3. Histograms of Numerical Features
4. Boxplots for Outlier Detection
5. Bar Charts for Categorical Variables
6. Countplots with Churn Comparison
7. Correlation Heatmap
8. Scatter Plot of Numerical Features

7. Key Findings

1. The dataset contains both numerical and categorical customer attributes.
2. Missing values and duplicate records were successfully handled.
3. Churn distribution revealed that a significant proportion of customers left the service.
4. Certain customer groups showed higher churn rates than others.

5. Numerical variables such as tenure and charges demonstrated noticeable differences between churned and non-churned customers.
6. Correlation analysis revealed relationships among several numerical features.
7. Outlier analysis identified extreme customer behavior patterns.
8. Visualizations provided insights into factors affecting customer retention.

8. Conclusion

The Customer Churn Analysis project successfully explored and cleaned the dataset, identified important patterns, and generated meaningful visualizations. The analysis revealed valuable insights into customer behavior and churn tendencies. The cleaned dataset is now suitable for machine learning models such as Logistic Regression, Random Forest, Decision Tree, XGBoost, or other predictive techniques to forecast future customer churn.

9. Future Scope

- Build churn prediction models.
- Perform feature engineering.
- Compare multiple machine learning algorithms.
- Deploy the model using Streamlit.
- Develop an interactive customer retention dashboard.